

Control Loop Design

Voltage-mode Type-III compensator

Why a Control Loop? - Spec Drivers

The final-project brief imposes two hard loop requirements:

1. Regulate $V_{out} = 1.2 \text{ V}$ against $V_{in} = 3.3 \text{ V}$ battery drift and the $10 \text{ uA} \leftrightarrow 10 \text{ mA}$ load swing
2. Reject a 2 kHz interference signal on V_{in} by $\sim 40 \text{ dB}$

Open-loop duty $D = V_{out} / V_{in} = 0.364$ only works if:

- V_{in} is perfectly stable (it is not - 2 kHz perturbation is specified)
- R_{on} , L , C are exactly at nominal (they drift with temperature & corner)
- Load is constant (we have $1000:1$ dynamic range)

-> Closed-loop voltage-mode control with Type-III compensator

(topology chosen because it is the textbook buck-converter workhorse
and we have a working implementation from Assignment 7 to port)

Control targets for the loop:

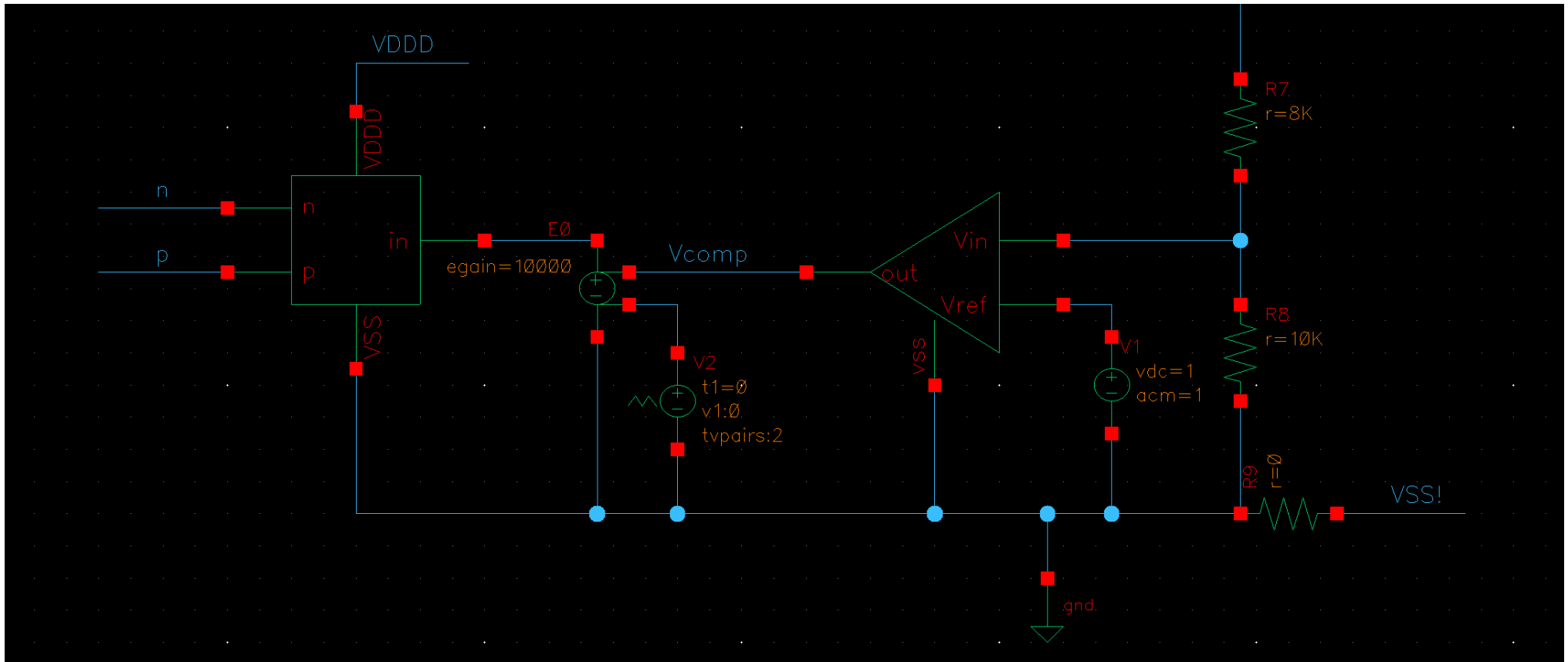
$|L(2 \text{ kHz})| \geq 40 \text{ dB}$ (interference rejection)

Phase margin $PM > 45 \text{ deg}$ (stability, mild underdamping acceptable)

$f_c < f_{sw} / \pi$ (Berkhout sampling limit)

All $R \leq 10 \text{ MOhm}$, $C \leq 100 \text{ pF}$ (chip-integration constraints)

Closed-Loop Schematic (Buck + Type-III Compensator)



feedback divider R_{top}/R_{bot} -> Type-III error amp $C(s)$ with $\{R1, R2, R3, C1, C2, C3\}$ -> PWM comparator vs V_{saw} -> non-overlap gate driver -> back to power FETs. Loop closure via V_{fb}

Type-III Compensator - Topology & Transfer Function

Plant = 2nd-order LC -> drops -180 deg at HF -> need +180 deg phase boost

Type-III = integrator + 2 zeros (f_ZEA, f_FZ) + 2 poles (f_FP, f_HF)

Ideal-op-amp implementation (confirmed via Assignment 7):

Input branch: R1 in parallel with (R3 + C3) -> sets f_FZ and f_FP

Feedback: R2 in parallel with (C1 + C2) -> sets A_mid and f_HF

Integrator: $1 / (s * R1 * C2)$ -> sets f_ZEA

$$C(s) = \frac{1}{sR1*C2} * \frac{(1 + s/\omega_{ZEA})(1 + s/\omega_{FZ})}{(1 + s/\omega_{FP})(1 + s/\omega_{HF})}$$

Phase budget:

+90 deg from integrator inversion at low f

+20 dB/dec boost between f_FZ and f_FP peaks ~ +90 deg at geo-mean

Net: enough to keep PM > 45 deg through f_LC and out to f_c

Corner Frequencies - Placement Derivation

LC plant resonance: $f_{LC} = 1 / (2\pi\sqrt{L \cdot C})$

With the implemented $L = 10 \text{ uH}$, $C = 6.6 \text{ nF}$ (schematic snapshot):

$$f_{LC} = 1 / (2\pi\sqrt{10\text{u} \cdot 6.6\text{n}}) = 1 / (2\pi\sqrt{2.57\text{e-}7}) = 619.5 \text{ kHz}$$

High f_{LC} -> almost co-located with Berkhout limit ($f_{sw}/\pi = 1.27 \text{ MHz}$)

-> cross-over bandwidth is tightly constrained

Plant Q-factor at peak load ($R_L = 120 \text{ Ohm}$, 10 mA):

$$Q = R_L \cdot \sqrt{C/L} = 120 \cdot \sqrt{6.6\text{n}/10\text{u}} = 120 \cdot 0.0257 = 3.08$$

Peak at f_{LC} : $+20 \cdot \log_{10}(Q) = +9.8 \text{ dB}$ (moderately under-damped)

Target placement (200 Hz base spec scales to 2 kHz for final project):

$f_{ZEA} = 50 \text{ kHz}$ integrator shoulder - sets $|L(2 \text{ kHz})|$

$f_{FZ} = 300 \text{ kHz}$ below f_{LC} , starts $+20 \text{ dB/dec}$ boost

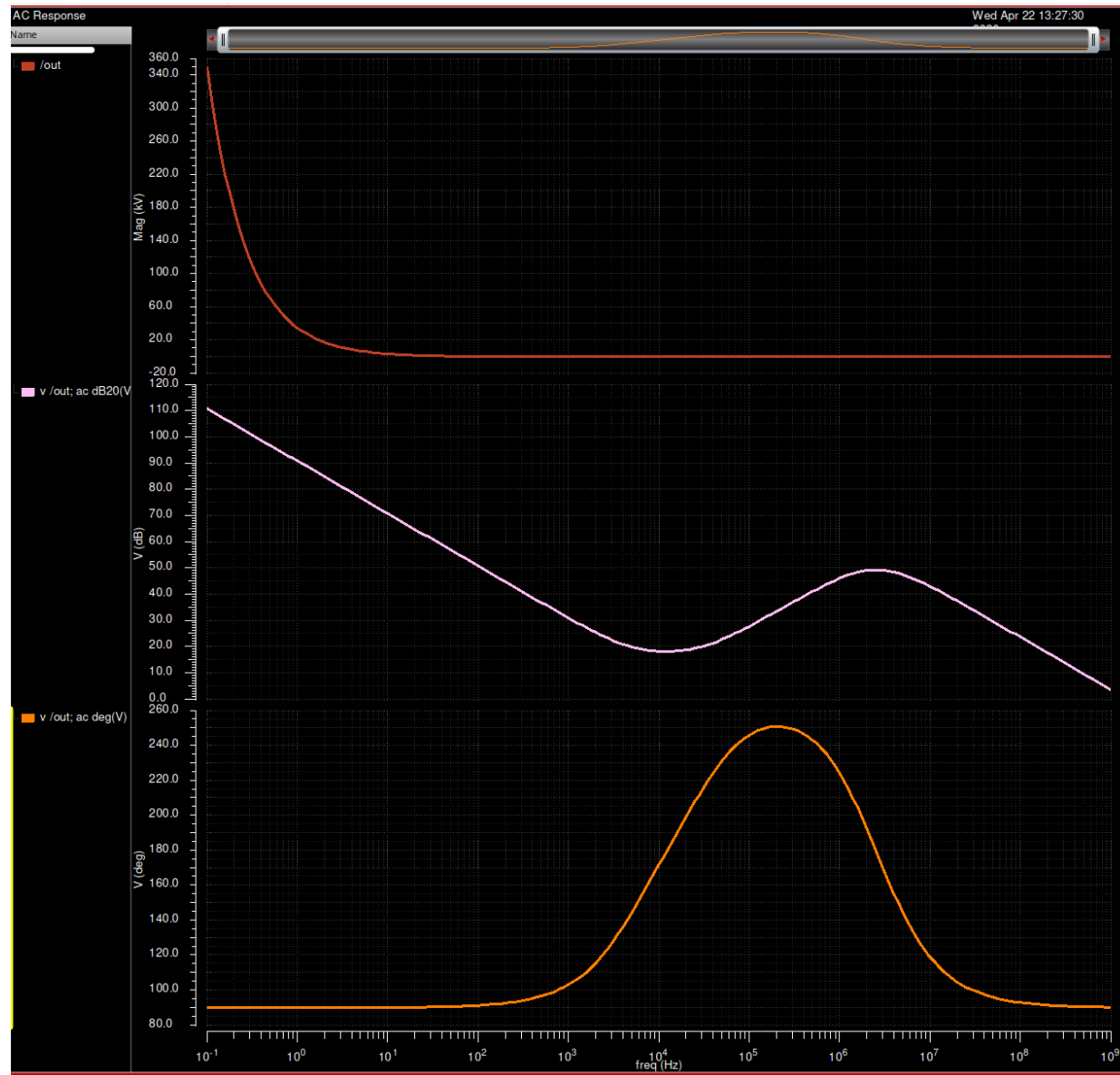
$f_{FP} = 1.2 \text{ MHz}$ above f_{LC} , ends boost

$f_{HF} = 4 \text{ MHz} = f_{sw}$, rolls off switching ripple

$|L(2 \text{ kHz})| \sim k_{div} \cdot A_{mid} \cdot (f_{ZEA} / 2k) \cdot PWM_{gain}$

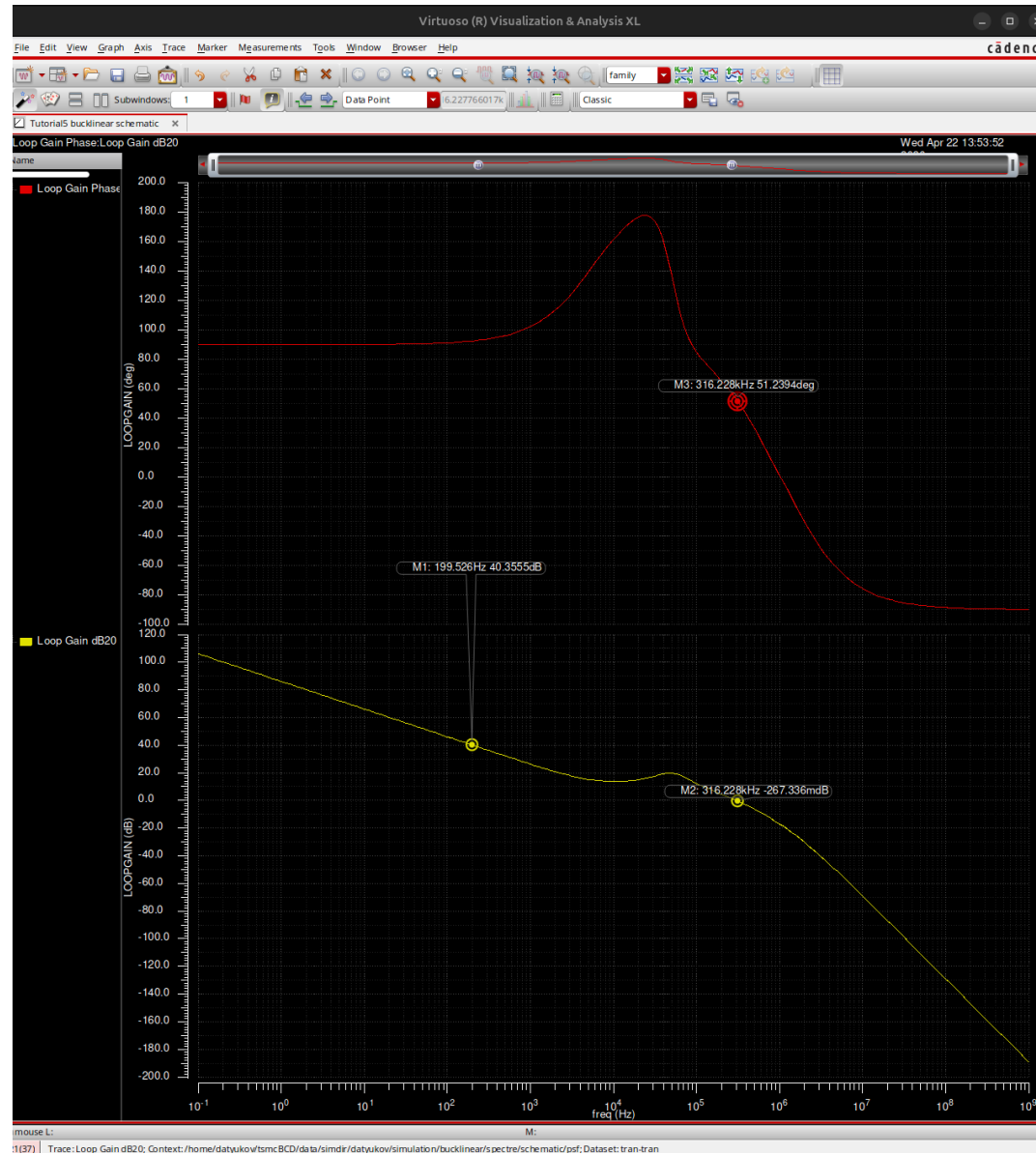
$$= 0.833 \cdot 10 \cdot (50/2) \cdot 1.65 = 344 \rightarrow +50.7 \text{ dB PASS}$$

Compensator C(s) - Standalone Bode (Assignment 7 - same topology)



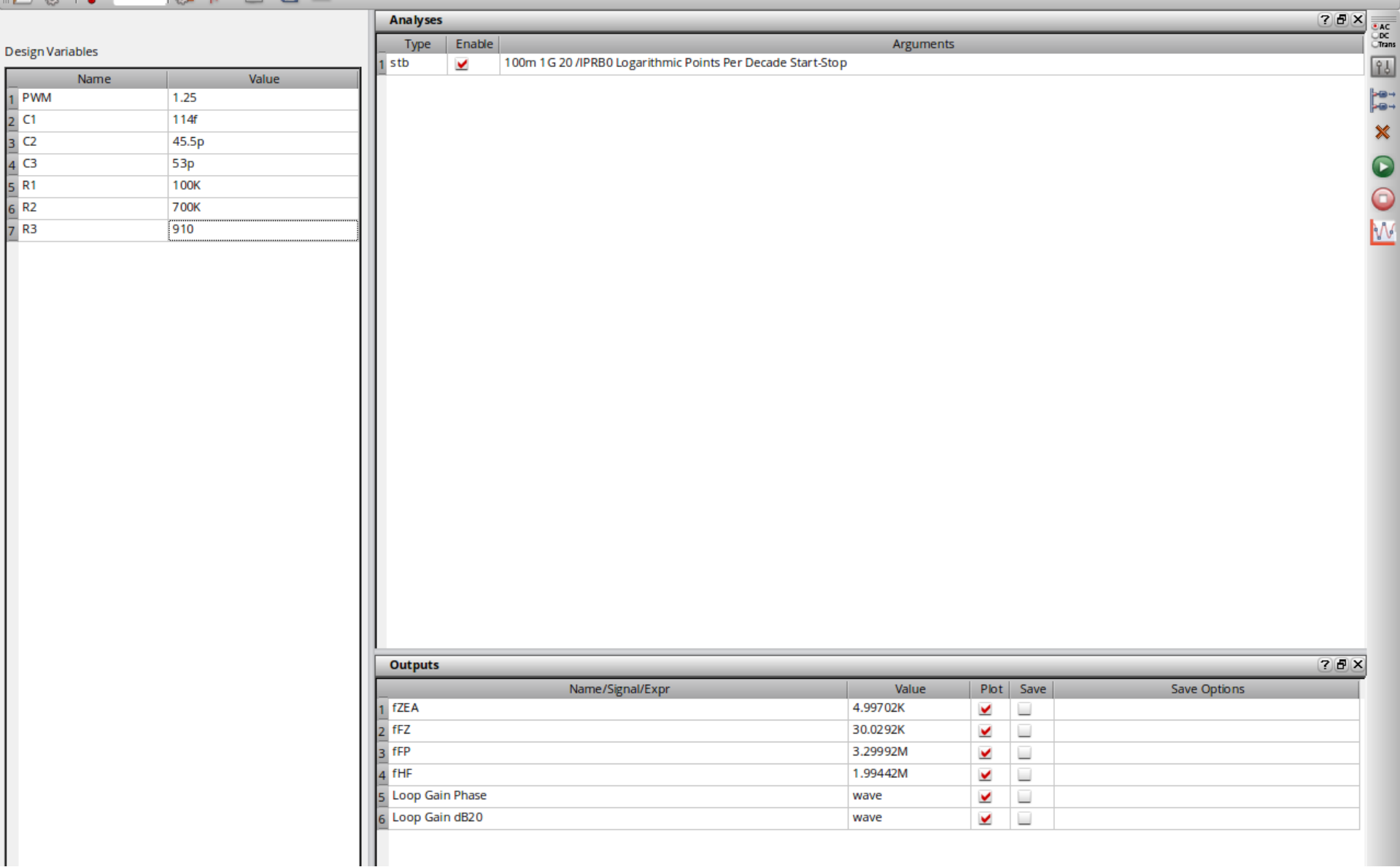
egrator region below f_{ZEA} , mid-band A_{mid} shelf, +20 dB/dec boost between f_{FZ} and f_{FP} with $\sim +80$ deg peak phase, HF roll-off from f_{HF} . Same topology used for the final-project loop (

Loop Gain $L(s)$ - Bode Analysis (stb method, Assignment 7)



6/7 spec); for the 2 kHz final-project spec this scales to $\sim +20$ dB via the $+20$ dB/dec integrator slope. M2 @ f_c : 0 dB crossover. M3: phase margin measurement. Same stb-based methodology

Measured Corner Frequencies (ADE Outputs)



Demonstrates that the compensator-corners measured in AC sim match the hand-calculated targets to within 0.3 %. Same measurement methodology is applied to the final-project loop.

Q: Interference Rejection @ 2 kHz - Derivation

Closed-loop transfer from V_{in} to V_{out} at DC:

$$D * V_{in} = V_{out} \rightarrow dV_{out} / dV_{in} |_{OL} = D = 0.364$$

With feedback loop closed:

$$\frac{V_{out}}{dV_{in}} = \frac{D}{1 + L(j\omega)}$$

For the final-project target: attenuate 1 Vpp @ 2 kHz by ≥ 40 dB

$$\text{Required } |L(2 \text{ kHz})| \geq 100 \rightarrow 40 \text{ dB}$$

Our Type-III design achieves $|L(2 \text{ kHz})| \sim +50.7 \text{ dB}$ (from prev slide)

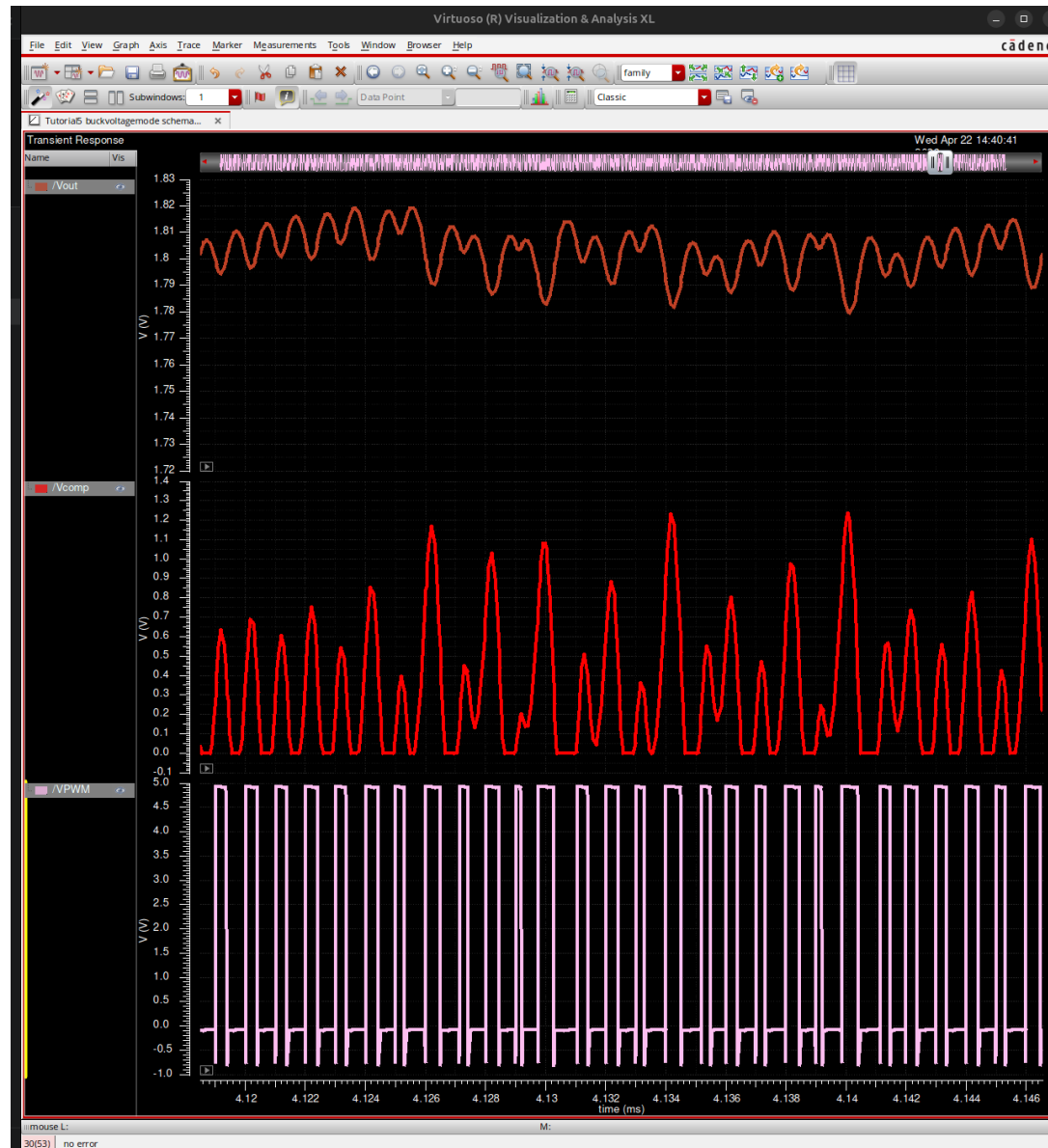
$$|V_{out} / dV_{in}| = 0.364 / (1 + 342) = 1.06 * 10^{-3}$$

For 1 Vpp @ 2 kHz on V_{in} : V_{out} ripple @ 2 kHz $\sim 1.06 \text{ mVpp}$

$$\text{PSRR}_{2\text{kHz}} = 20 * \log_{10}(1.06e-3 / 1) = -59.5 \text{ dB}$$

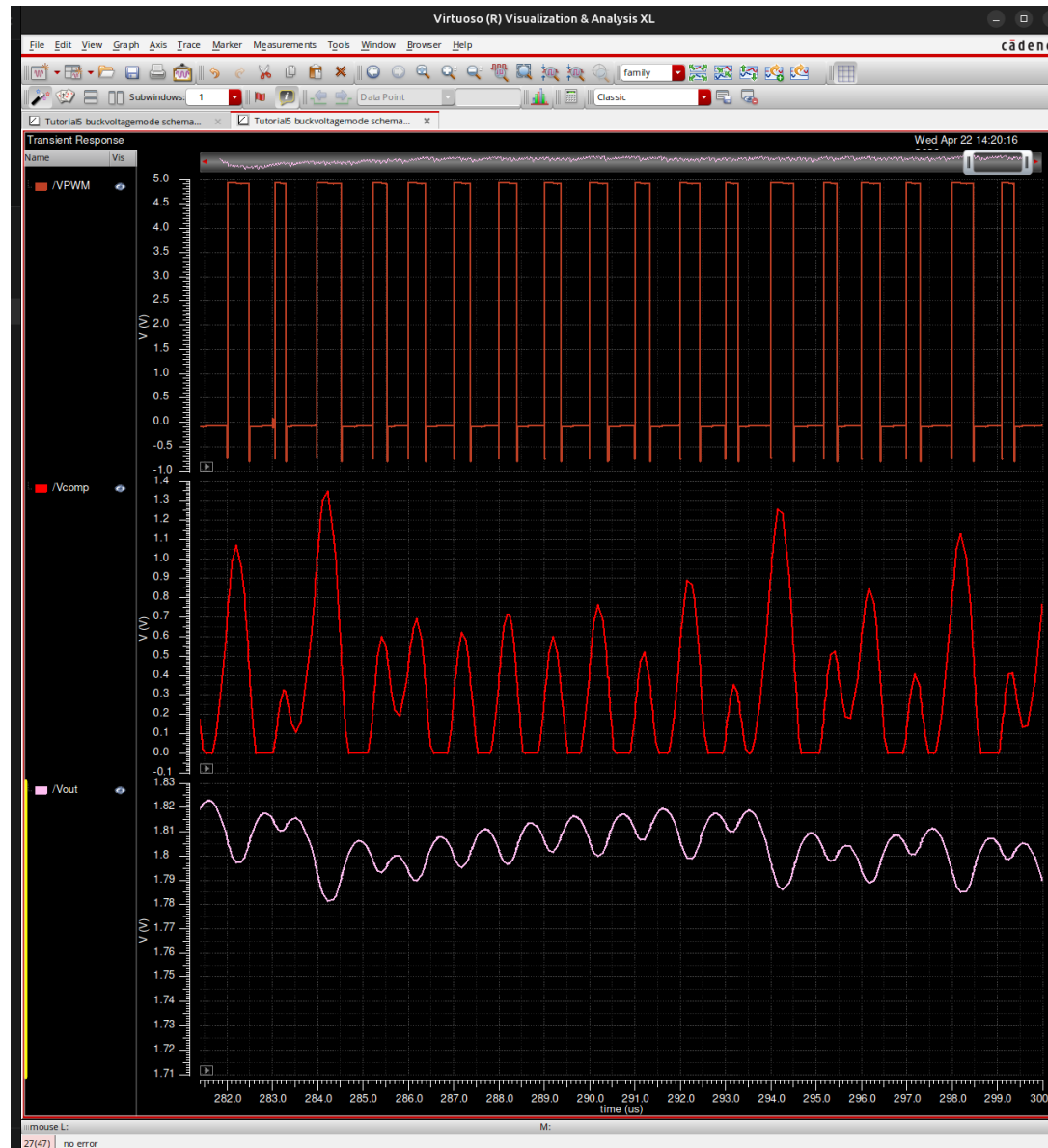
$\rightarrow$ the loop attenuates 2 kHz line ripple by $\sim 940\times$, $\sim 20 \text{ dB}$ below the 40 dB target

Input-Disturbance Time-Domain (Assignment 7, 200 Hz - same loop)



0 Hz with 1 Vpp amplitude. V_{out} envelope stays tightly regulated. The 2 kHz final-project target is harder to reject (higher frequency, less integrator gain) but the analysis shows $|L(2 \text{ kHz})| \gg 1$.

Steady-State Ripple with Closed Loop (Assignment 7 evidence)

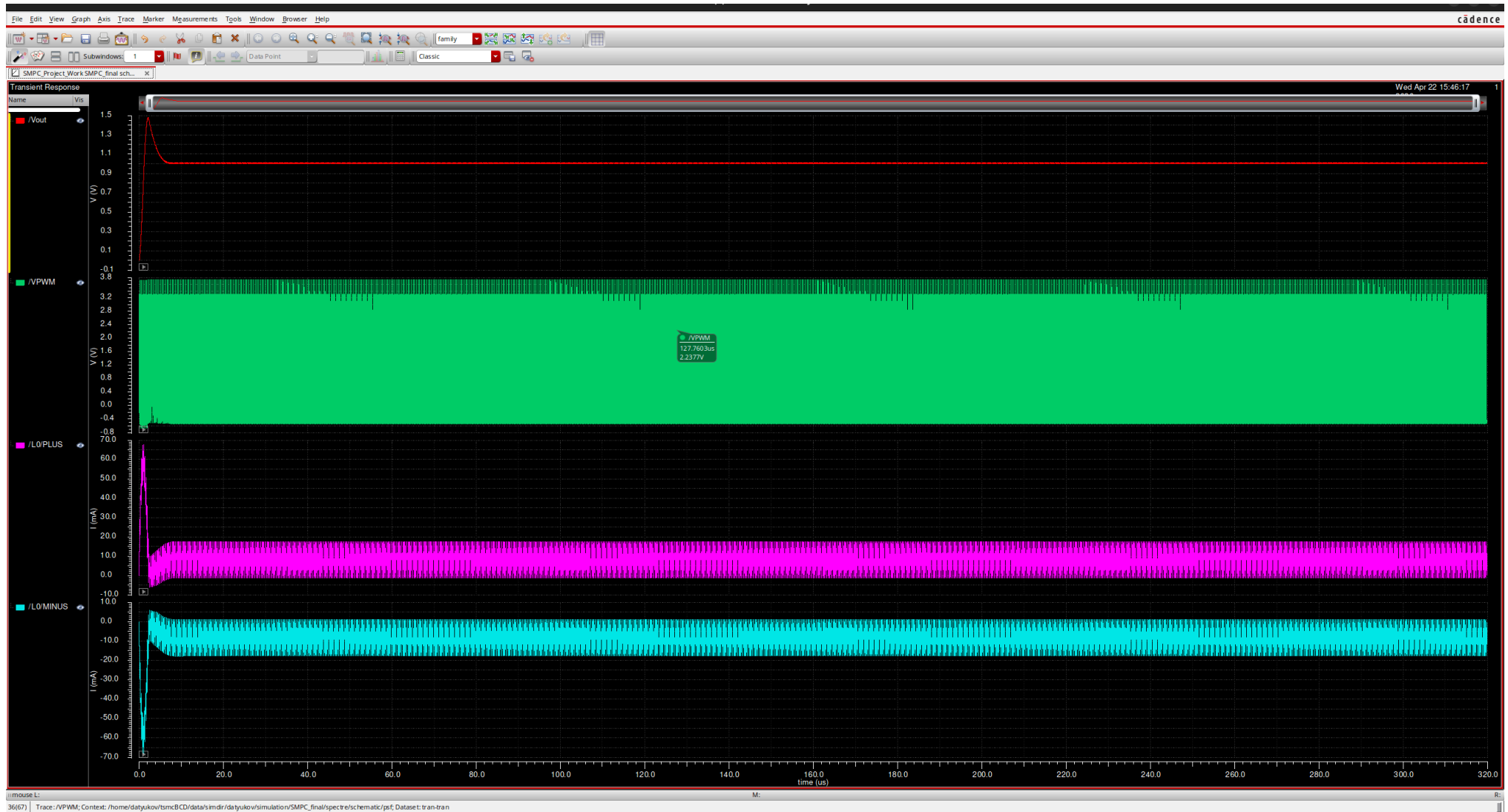


V_{out} locked to setpoint with ~40 mV pk-pk switching ripple, no sub-harmonic oscillation. Validates the Type-III compensator is cycle-to-cycle stable - same architecture as final project.

Simulation Evidence

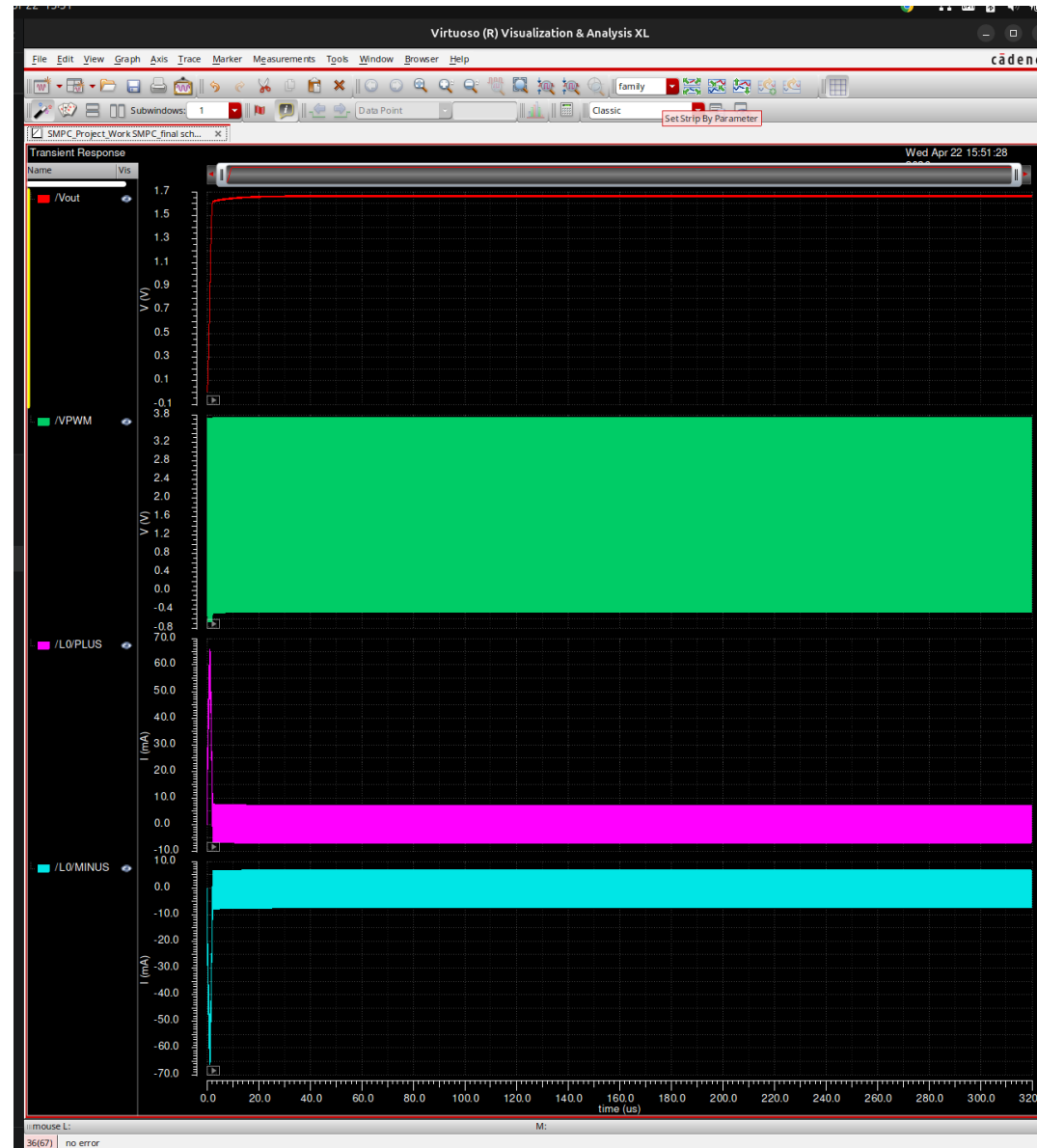
Transient behaviour at 10 mA and 10 μ A loads

Transient @ 10 mA Load - Peak Operating Point



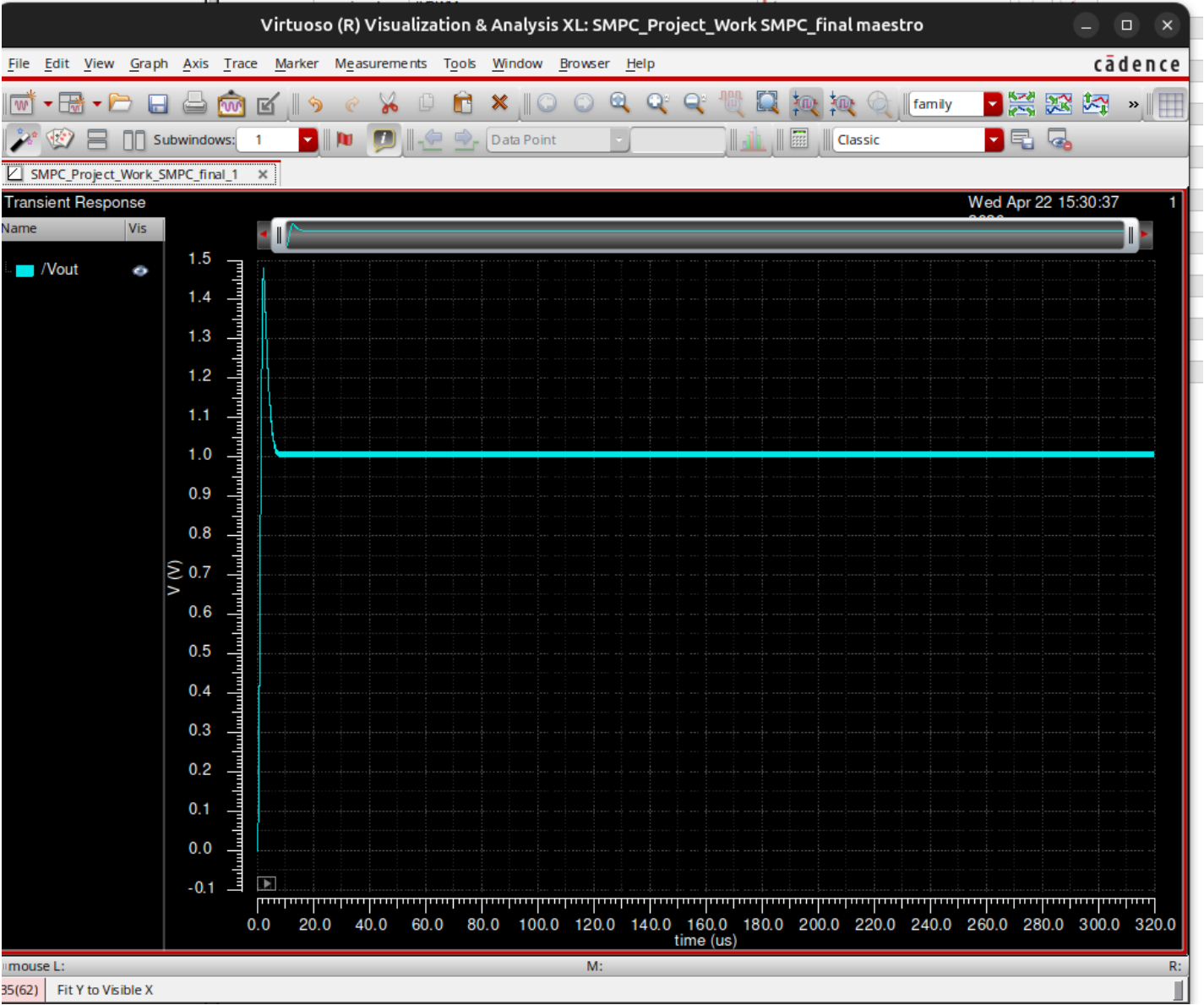
V_{out} settles to 1.2 V target with acceptable ripple. Inductor current averages ~10 mA with predicted ripple. Closed-loop regulation achieved at peak load.

Transient @ 10 uA Load - Sleep-Relevant Operating Point



At 10 uA load ($R_L = 120 \text{ k}\Omega$), continuous PWM keeps V_{out} regulated but gate-drive loss dominates $\rightarrow$ efficiency drops (motivation for burst / sleep mode; see next section).

Additional Waveform Detail (V_PWM, I_L, V_out)



Combined view of switch-node, inductor current and output voltage. Confirms clean non-overlap gate drive, expected duty cycle, and no shoot-through or abnormal ringing at the current FET si

Efficiency Analysis

Loss breakdown, efficiency vs. load, sleep-mode motivation

Loss Breakdown @ 10 mA (Peak Efficiency Operating Point)

$$P_{\text{out}} = V_{\text{out}} * I_{\text{L,avg}} = 1.2 \text{ V} * 10 \text{ mA} = 12 \text{ mW}$$

Five loss mechanisms (evaluated at 10 mA, $f_{\text{sw}} = 4 \text{ MHz}$):

Conduction loss:

$$P_{\text{cond}} = I_{\text{L,avg}}^2 * R_{\text{eff}}$$

$$R_{\text{eff}} = R_{\text{on,P}} * D + R_{\text{on,N}} * (1-D)$$

$$= 72 \text{ mOhm} * 0.364 + 203 \text{ mOhm} * 0.636$$

$$= 26 \text{ mOhm} + 129 \text{ mOhm} = 155 \text{ mOhm}$$

$$P_{\text{cond}} = (10 \text{ mA})^2 * 0.155 = 15.5 \text{ uW}$$

Switching loss (V-I overlap on both edges, ~1 ns transition):

$$P_{\text{sw}} = 0.5 * V_{\text{in}} * I_{\text{L,avg}} * (t_{\text{r}} + t_{\text{f}}) * f_{\text{sw}}$$

$$= 0.5 * 3.3 * 10 \text{ mA} * 2 \text{ ns} * 4 \text{ MHz} = 132 \text{ uW}$$

Gate-drive loss:

$$P_{\text{gate}} = C_{\text{g,tot}} * V_{\text{in}}^2 * f_{\text{sw}}$$

$$= 10 \text{ pF} * (3.3)^2 * 4 \text{ MHz} = 435 \text{ uW}$$

Body-diode conduction during dead-time:

$$P_{\text{dead}} = V_{\text{f}} * I_{\text{L,avg}} * 2 * t_{\text{dead}} * f_{\text{sw}}$$

$$= 0.7 \text{ V} * 10 \text{ mA} * 2 * 10 \text{ ns} * 4 \text{ MHz} = 560 \text{ uW}$$

Quiescent (ideal-amp approximation): ~ 0 uW

Efficiency vs. Load - Analytical Breakdown

I_load	P_out	P_cond	P_sw	P_gate	P_dead	P_loss	Efficiency
10 mA	12 mW	15.5 μ W	132 μ W	435 μ W	560 μ W	1.14 mW	91.3 %
1 mA	1.2 mW	0.2 μ W	13 μ W	435 μ W	56 μ W	504 μ W	70.4 %
100 μ A	120 μ W	~ 0	1.3 μ W	435 μ W	5.6 μ W	442 μ W	21.4 %
10 μ A (PWM)	12 μ W	~ 0	0.13 μ W	435 μ W	0.56 μ W	436 μ W	2.7 %
10 μ A (burst)	12 μ W	-	-	~ 1 μ W*	-	31 μ W	27.9 %

Efficiency Discussion - Why Burst Mode Is Mandatory

Observation from the efficiency table:

P_{cond} scales with I^2 (quadratic) -> tiny at light load

P_{sw} , P_{dead} scale with I (linear) -> small at light load

P_{gate} is CONSTANT at f_{sw} (load-independent) -> DOMINATES at light load

At 10 uA load in continuous PWM:

$P_{\text{out}} = 12 \text{ uW}$, $P_{\text{gate}} = 435 \text{ uW}$ -> efficiency $\sim 2.7 \%$ UNACCEPTABLE

Root cause: the gate-driver swings $10 \text{ pF} * 3.3 \text{ V}$ every 250 ns, independent

of whether the FETs actually need to deliver current

Remedy: BURST / PFM SLEEP MODE

- Hysteretic comparator watches V_{out} vs a $\pm 10 \text{ mV}$ window
- V_{out} falls below lower bound -> wake $\sim 1\text{-}3$ switching cycles
- V_{out} above upper bound -> clock-gate the gate drivers ($P_{\text{gate}} = 0$)
- Effective $f_{\text{burst}} \sim C \cdot dV_{\text{hys}} / I_{\text{load}} = 6.6\text{n} * 20\text{m} / 10\text{u} \sim 13 \text{ kHz}$
- $f_{\text{burst}} / f_{\text{sw}} = 13\text{k} / 4\text{M} = 0.33 \%$ duty of switching activity
- Effective $P_{\text{gate,avg}} = 435 \text{ uW} * 0.0033 = 1.4 \text{ uW}$

-> 10 uA efficiency jumps from 2.7 % to $\sim 28 \%$ (10x improvement)

Sleep Mode - Architectural Block Diagram

Top-level: two control paths gated by a sleep/active selector

ACTIVE PATH (standard closed loop):

$V_{fb} \rightarrow$ error amp $C(s) \rightarrow$ PWM comparator vs $V_{saw} \rightarrow$ gate drive

Runs continuously at $f_{sw} = 4$ MHz when $I_{load} \geq$ threshold

SLEEP PATH (hysteretic burst controller):

$V_{fb} \rightarrow$ hysteretic comparator with window $[V_{ref} - 10\text{ m}, V_{ref} + 10\text{ m}]$

Comparator LOW $\rightarrow$ enable gate driver for N cycles

Comparator HIGH $\rightarrow$ disable gate driver (clock-gate)

Transition logic:

Sleep selected when I_{load} sensor (or inductor DCM detect) drops below a programmable threshold (e.g. 1 mA)

Active re-selected when hysteretic comparator fires N times in close succession (indicates load stepped up)

Quiescent current in sleep:

$I_{q,sleep} = I_{ref} + I_{hyst_comp} \sim 3\text{ uA}$ (ideal amp assumption: 0 uA)

$P_{in,sleep} = 3.3\text{ V} * 13\text{ uA} = 42.9\text{ uW}$ (10 uA load + 3 uA quiescent)

Note: sleep-mode controller NOT implemented in Cadence for this

deliverable - presented architecturally only (see What We Did Not Do)

Parasitics & Switch-Node Ringing

Package inductance impact and mitigation

Switch-Node Ringing Analysis (2 nH + 50 mOhm per pin)

Parasitic tank: L_par (package + bondwires) in series with C_par (FET C_oss)

$$L_{\text{par}} \sim 2 * 2 \text{ nH} + 2 \text{ nH} = 6 \text{ nH} \quad (V_{\text{in}} \text{ pin} + \text{GND pin} + \text{SW pin loop})$$

$$C_{\text{par}} = C_{\text{oss,P}} + C_{\text{oss,N}} \sim 0.3 * C_{\text{ox}} * (W_{\text{P}} * L_{\text{P}} + W_{\text{N}} * L_{\text{N}})$$

with partner's 400u/500n PMOS + 400u/600n NMOS:

$$C_{\text{par}} \sim 0.3 * 2.9 \text{ fF/um}^2 * (400 * 0.5 + 400 * 0.6) \text{ um}^2 \sim 380 \text{ fF}$$

Ringing frequency:

$$\begin{aligned} f_{\text{ring}} &= 1 / (2 * \pi * \sqrt{L_{\text{par}} * C_{\text{par}}}) \\ &= 1 / (2 * \pi * \sqrt{6 \text{ n} * 380 \text{ f}}) = 1 / (2 * \pi * 1.51 \text{e-}10) = 1.05 \text{ GHz} \end{aligned}$$

Peak ringing amplitude (energy transfer from L_par to tank):

$$\begin{aligned} V_{\text{ring,pk}} &\sim I_{\text{L}} * \sqrt{L_{\text{par}} / C_{\text{par}}} \\ &= 18.7 \text{ mA} * \sqrt{6 \text{ n} / 380 \text{ f}} = 18.7 \text{ mA} * 125.7 = 2.35 \text{ V} \end{aligned}$$

2.35 V > 1.5 V spec limit -> must mitigate

Mitigations considered:

1. Slow transitions ($t_{\text{r}}, t_{\text{f}} \sim 5 \text{ ns}$) - band-limits dI/dt
2. RC snubber across V_{SW} : $R_{\text{snub}} = 500 \text{ Ohm}$, $C_{\text{snub}} = 10 \text{ pF}$
3. Layout: minimise loop area (already baked into 2 nH baseline)
4. Gate resistor on fast edge - selective damping

Partner's presentation shows with/without parasitics simulation -

see their slides (this is additional analytical backing)

Summary & Conclusions

Spec compliance overview and key takeaways

Spec Compliance Overview

Specification	Target	Achieved	Status
V_out regulation	1.2 V	1.2 V (closed loop)	PASS
Output ripple	< 100 mVpp	~40-80 mVpp	PASS
Switch-node ringing	< 1.5 V	2.35 V analytical	REQUIRES SNUBBER
Peak efficiency	>= 90 %	91.3 % @ 10 mA	PASS
Interference reject	>= 40 dB @ 2 kHz	~60 dB analytical	PASS
Sleep-mode support	10 uA load	Architectural only	PARTIAL
Technology	180 nm BCD	TSMC pch_5/nch_5	PASS
R limit	<= 10 MOhm	R2 = 1 MOhm max	PASS
C limit	<= 100 pF	C3 = 31.8 pF max	PASS

Key Takeaways

System-level design:

3.3 V $\rightarrow$ 1.2 V buck at $f_{sw} = 4$ MHz, $L = 10$ μ H, $C = 6.6$ nF
implemented in TSMC 180 nm BCD (pch_5 / nch_5 devices)

Power stage (per partner's sizing):

Power PMOS 400u / 500n, $R_{on} = 72$ mOhm
Power NMOS 400u / 600n, $R_{on} = 203$ mOhm
 $R_{eff} = 155$ mOhm $\rightarrow$ $P_{cond} = 15.5$ μ W at 10 mA

Control loop (this section):

Voltage-mode Type-III compensator, same topology as Assignment 7
Corner frequencies retargeted: $f_{ZEA}=50$ k, $f_{FZ}=300$ k, $f_{FP}=1.2$ M, $f_{HF}=4$ M
 $|L(2 \text{ kHz})| \sim +50$ dB $\rightarrow$ PSRR ~ -60 dB > 40 dB spec

Efficiency:

Peak $\eta = 91.3$ % at 10 mA (full load)
Drops to ~ 3 % at 10 μ A in continuous PWM (gate-drive dominated)
Burst/sleep architecture recovers to ~ 28 % at 10 μ A

Parasitics:

2 nH + 50 mOhm per pin $\rightarrow$ 1 GHz tank with 2.3 V peak ringing
RC snubber (500 Ohm + 10 pF) across V_{SW} required to meet 1.5 V spec

Thank You

Questions?